CHANDIGARH UNIVERSITY

Discover. Learn. Empower.

Subject: DAMS

Subject Code: 24CSH-298

Submitted By:

Name: Utkarsh

UID No: 24BCF10020

Branch/Sec: 24AIT-KRG-G2

Submitted To:

Faculty Name: Mr. Shalabh Bhatiya

Q1) Explain the following types of database security issues: Legal and Ethical issues, Policy issues and System related Issues.

Ans Database security issues arise when there are threats to the confidentiality, integrity and availability of data. These issues can be categorized as legal and ethical issues, policy issues, and system-related issues.

1) Legal and Ethical Issues

Legal and ethical concern the laws, regulations, and moral responsibilities related to the storage and use of data in a database system.

Organizations must ensure that they comply with data protection laws and privacy regulations. Sensitive information such as personal data, financial records, and confidential business information should not be accessed or shared without proper authorization.

Example:- Leaking a customer personal information or using employee data without consent.

2) Policy Issues

Policy issues relate to the rules and guidelines established by an organization to protect it's database systems. Organizations develop security policies that define how data should be accessed, used, and protected. These policies specify user roles, access rights, responsibilities of employees when handling data.

3) System-related Issues

System-related issues arise due to technical weak-ness or vulnerbilities in database systems, hardware, or software.

These issues are may include weak authentication mechanisms, software bugs, improper system configuration, or cyber attacks such as hacking and malware. If the system is not properly secured, attackers may gain unauthorized acces modify data or distrupt database operations.

Q2) Exchlain the following database security control techniques: 1) Access Control 2) Inte-ference Control 3) Flow Control 4) Datam-cryption.

Ans Database security control techniques are meat-hods used to protect databases from unauthori-zed access, misuse, or data leakage.

1) Access Control :- Access Control is a technique used to restrict database access to authorized access to authorized users only. It determines who can access the database and what operation they are allowed to perform. Access control is implemented through user authentication, roles and permission which can be granted by data administrators which can be granted and revoked.

Example :- A user may be given permission to view employee records but not modify them.

DeptID). write a SQL query to create a

2) Interference Control:- Inference Control is used to prevent users from deducing sensitive information from non-sensitive data. Sensitive information can sometimes be accessed by users by analyzing available data or statistical results. These control techniques restrict certain queries or limit the results returned to avoid such situations.

Example:- a user may not be allowed to view individual salaries but could try to calculate them using average data.

3) Flow Control :- Flow Control regulates the movement of information within a database system. It ensures that information flows only between authorized users and security levels. This technique prevents data from moving from higher security level to lower security level where unauthorized users may access it.

4) Data encryption :- Data encryption is the process of converting readable data into an encoded format to protect it from unauthorized access.

Encrypt data can only be read by users who have the correct decryption key. encryption commonly used to protect sensitive data during storage transfer over networks.

Q3) Consider a table Employee(EmpID, EmpName, Salary, DeptID). Write a SQL query to create a

view that displays only EmpID, EmpName, and DeptID from the employee table.

Ans. A view in SQL is a virtual that displays data from one or more tables based on a query

To create a view that shows only EmpID, EmpName and DeptID from the employee table, the following SQL query can be used.

"CREATE VIEW Employee-View AS
SELECT EmpID, EmpName., DeptID
FROM Employee;"

This view allows users to access selected colums while hiding Salary Colum from the original table

Description | Editorial | Solutions | Submissions

577. Employee Bonus

Solved

Easy | Topics | Companies | Hint

SQL Schema > Pandas Schema >

Table: `Employee`

```
+-------------+---------+
| Column Name | Type    |
+-------------+---------+
| empId       | int     |
| name        | varchar |
| supervisor  | int     |
| salary      | int     |
+-------------+---------+
```

empId is the column with unique values for this table.
Each row of this table indicates the name and the ID of an employee in addition to their salary and the id of their manager.

Table: `Bonus`

```
+-------------+------+
| Column Name | Type |
```

1.6K | 221 | 45 Online

Code

MySQL | Auto

```mysql
SELECT e.name , b.bonus
FROM employee e
LEFT JOIN bonus b ON e.empId = b.empID
WHERE bonus IS NULL OR bonus < 1000;
```

Saved | Ln 1, Col 1

Accepted

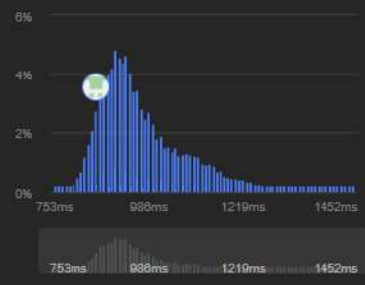
All Submissions

Accepted 21 / 21 testcases passed

uk-06 submitted at Feb 28, 2026 22:20

Runtime

856 ms | Beats **92.39%**



Code | MySQL

Testcase | Test Result

Description | Editorial | Solutions

570. Managers with at Least 5 Direct Reports

Solved

Medium | Topics | Companies | Hint

SQL Schema > Pandas Schema >

Table: `Employee`

```
+-------------+---------+
| Column Name | Type    |
+-------------+---------+
| id          | int     |
| name        | varchar |
| department  | varchar |
| managerId   | int     |
+-------------+---------+
```

id is the primary key (column with unique values) for this table.
Each row of this table indicates the name of an employee, their department, and the id of their manager.
If managerId is null, then the employee does not have a manager.
No employee will be the manager of themself.

1.8K | 367 | 49 Online

Code

MySQL | Auto

```sql
select e.name from employee e
inner join employee r on e.id = r.managerID
group by r.managerID having count(r.managerID) >=5
```

Saved | Ln 1, Col 1

Accepted

All Submissions

Accepted 11 / 11 testcases passed

uk-06 submitted at Mar 01, 2026 16:05

Runtime

330 ms | Beats 68.31%

Submissions

Testcase | Test Result